```
<!DOCTYPE html>
<html lang="ms">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Misi Detektif: Teka Alatan Keselamatan</title>
  <style>
    * {
      box-sizing: border-box;
    }
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: linear-gradient(135deg, #e0f2fe, #fef3c7);
      color: #243047;
      min-height: 100vh;
      display: flex;
      justify-content: center;
      align-items: center;
      padding: 20px;
    }
    .container {
      background: white;
      width: 100%;
```

```
  max-width: 650px;
  padding: 30px;
  border-radius: 25px;
  box-shadow: 0 10px 30px #00000015;
  text-align: center;
}
h1 {
  color: #1d4ed8;
  margin-bottom: 10px;
}
.subtitle {
  color: #64748b;
  line-height: 1.6;
}
.badge {
  display: inline-block;
  background: #dbeafe;
  color: #1d4ed8;
  padding: 8px 15px;
  border-radius: 20px;
  font-weight: bold;
  margin: 12px 0;
}
.progress {
```

```
  height: 10px;
  background: #e2e8f0;
  border-radius: 20px;
  overflow: hidden;
  margin: 20px 0;
}
.progress-bar {
  height: 100%;
  width: 0;
  background: #22c55e;
  transition: width 0.3s;
}
.picture {
  width: 100%;
  height: 270px;
  object-fit: contain;
  background: #f8fafc;
  border: 3px dashed #bfdbfe;
  border-radius: 20px;
  padding: 12px;
}
.question {
  font-size: 21px;
  font-weight: bold;
```

```
  margin: 22px 0;
}
input {
  width: 100%;
  padding: 14px;
  border: 2px solid #cbd5e1;
  border-radius: 12px;
  font-size: 16px;
  margin-bottom: 12px;
}
button {
  border: none;
  border-radius: 12px;
  padding: 13px 18px;
  margin: 5px;
  cursor: pointer;
  font-size: 15px;
  font-weight: bold;
  transition: 0.2s;
}
button:hover {
  transform: translateY(-2px);
  opacity: 0.9;
}
```

```
.primary {
  background: #2563eb;
  color: white;
}
.green {
  background: #16a34a;
  color: white;
}
.orange {
  background: #f59e0b;
  color: white;
}
.gray {
  background: #e2e8f0;
  color: #334155;
}
.feedback {
  padding: 15px;
  border-radius: 12px;
  margin: 15px 0;
  line-height: 1.6;
  display: none;
}
.correct {
```

```
  display: block;
  background: #dcfce7;
  color: #166534;
}
.wrong {
  display: block;
  background: #fee2e2;
  color: #991b1b;
}
.hidden {
  display: none;
}
.score {
  font-size: 25px;
  color: #16a34a;
  font-weight: bold;
}
.note {
  font-size: 13px;
  color: #64748b;
  line-height: 1.6;
  margin-top: 20px;
}
@media (max-width: 480px) {
```

```
      .container {
        padding: 20px 15px;
      }
      .picture {
        height: 220px;
      }
      button {
        width: 100%;
        margin: 5px 0;
      }
    }
  </style>
</head>
<body>
  <main class="container">
    <section id="intro">
      <div style="font-size:55px;">🕵️</div>
      <h1>Misi Detektif!</h1>
      <h2>Teka Alatan Keselamatan</h2>
      <p class="subtitle">
        Adakah kamu bersedia menjadi detektif keselamatan?
        Perhatikan gambar, teka nama alatan dan sebut jawapan
        kamu dengan kuat!
      </p>
```

```
  <p>🎯 Lihat gambar · 🗣️ Sebut jawapan · ⭐ Kumpul markah</p>
  <button class="primary" onclick="startQuiz()">
    Mula Kuiz 🚀
  </button>
</section>
<section id="quiz" class="hidden">
  <div class="badge" id="questionNumber">Soalan 1</div>
  <div class="progress">
    <div class="progress-bar" id="progressBar"></div>
  </div>
  <img id="toolImage" class="picture" src="" alt="Gambar alatan keselamatan">
  <p class="question">
    Apakah nama alatan keselamatan dalam gambar ini?
  </p>
  <p class="subtitle">
    Sebut jawapan kamu terlebih dahulu sebelum menyemaknya.
  </p>
  <button class="orange" onclick="speakAnswer()">
    🎤 Sebut Jawapan
  </button>
  <p id="voiceStatus" aria-live="polite"></p>
  <label for="answer"><strong>Atau taip jawapan kamu:</strong></label>
  <input
    id="answer"
```

```
    type="text"
    placeholder="Taip nama alatan di sini..."
    autocomplete="off"
  >
  <button class="primary" id="checkBtn" onclick="checkAnswer()">
    Semak Jawapan
  </button>
  <div id="feedback" class="feedback" aria-live="polite"></div>
  <button id="nextBtn" class="green hidden" onclick="nextQuestion()">
    Soalan Seterusnya ➜
  </button>
  <p class="note">
    Tip: Dengar arahan guru dan jawab dengan yakin!
  </p>
</section>
<section id="result" class="hidden">
  <div style="font-size:60px;">🏆</div>
  <h1>Tahniah, Detektif!</h1>
  <p class="subtitle">Kamu telah menyelesaikan semua soalan.</p>
  <p class="score" id="finalScore"></p>
  <p id="finalMessage"></p>
  <button class="primary" onclick="restartQuiz()">
    🔄 Cuba Lagi
  </button>
```

```
  </section>
</main>
<script>
  /* GANTIKAN URL GAMBAR DAN NAMA ALATAN DI BAWAH DENGAN GAMBAR SEBENAR DARIPADA VIDEO YOUTUBE. Contoh URL: images/topi-keselamatan.jpg */
  const questions = [
    {
      name: "Topi keselamatan",
      image: "images/topi-keselamatan.jpg",
      aliases: ["topi keselamatan", "helmet"],
      function: "Melindungi kepala daripada hentakan dan kecederaan."
    },
    {
      name: "Cermin mata keselamatan",
      image: "images/cermin-mata-keselamatan.jpg",
      aliases: ["cermin mata keselamatan", "goggle", "goggles"],
      function: "Melindungi mata daripada habuk, serpihan dan percikan tertentu."
    },
    {
      name: "Sarung tangan keselamatan",
      image: "images/sarung-tangan.jpg",
      aliases: ["sarung tangan keselamatan", "sarung tangan"],
      function: "Melindungi tangan daripada bahaya tertentu mengikut jenis sarung tangan."
    },
    {
```

```
      name: "Kasut keselamatan",
      image: "images/kasut-keselamatan.jpg",
      aliases: ["kasut keselamatan", "kasut safety"],
      function: "Melindungi kaki daripada risiko seperti hentakan atau objek berat, bergantung pada jenis kasut."
    },
    {
      name: "Pelindung telinga",
      image: "images/pelindung-telinga.jpg",
      aliases: ["pelindung telinga", "penutup telinga", "ear muff", "earmuff"],
      function: "Mengurangkan pendedahan kepada bunyi bising apabila digunakan dengan betul."
    }
  ];
  let currentQuestion = 0;
  let score = 0;
  let answered = false;
  let recognition = null;
  function startQuiz() {
    document.getElementById("intro").classList.add("hidden");
    document.getElementById("result").classList.add("hidden");
    document.getElementById("quiz").classList.remove("hidden");
    currentQuestion = 0;
    score = 0;
    showQuestion();
```

```
}
function showQuestion() {
  answered = false;
  const q = questions[currentQuestion];
  document.getElementById("questionNumber").textContent =
    `Soalan ${currentQuestion + 1} daripada ${questions.length}`;
  document.getElementById("progressBar").style.width =
    `${(currentQuestion / questions.length) * 100}%`;
  document.getElementById("toolImage").src = q.image;
  document.getElementById("toolImage").alt =
    "Gambar alatan keselamatan untuk diteka";
  document.getElementById("answer").value = "";
  document.getElementById("answer").disabled = false;
  document.getElementById("feedback").className = "feedback";
  document.getElementById("feedback").textContent = "";
  document.getElementById("voiceStatus").textContent = "";
  document.getElementById("checkBtn").classList.remove("hidden");
  document.getElementById("checkBtn").disabled = false;
  document.getElementById("nextBtn").classList.add("hidden");
}
function normalize(text) {
  return text
    .toLocaleLowerCase("ms")
    .normalize("NFD")
```

```
    .replace(/[\u0300-\u036f]/g, "")
    .replace(/[^\p{L}\p{N}\s]/gu, "")
    .trim()
    .replace(/\s+/g, " ");
}
function checkAnswer() {
  if (answered) return;
  const userAnswer = normalize(
    document.getElementById("answer").value
  );
  if (!userAnswer) {
    document.getElementById("feedback").className = "feedback wrong";
    document.getElementById("feedback").textContent =
      "Sila masukkan jawapan dahulu atau gunakan mikrofon.";
    return;
  }
  const q = questions[currentQuestion];
  const acceptedAnswers = [q.name, ...q.aliases].map(normalize);
  const isCorrect = acceptedAnswers.includes(userAnswer);
  answered = true;
  const feedback = document.getElementById("feedback");
  if (isCorrect) {
    score++;
    feedback.className = "feedback correct";
```

```
    feedback.innerHTML =
      `🎉 Syabas! Jawapan kamu betul!<br>
       <strong>${q.name}</strong><br>
       Fungsi: ${q.function}`;
  } else {
    feedback.className = "feedback wrong";
    feedback.innerHTML =
      `Jangan putus asa!<br>
       Jawapan yang betul ialah <strong>${q.name}</strong>.<br>
       Fungsi: ${q.function}`;
  }
  document.getElementById("answer").disabled = true;
  document.getElementById("checkBtn").classList.add("hidden");
  document.getElementById("nextBtn").classList.remove("hidden");
  document.getElementById("nextBtn").textContent =
    currentQuestion === questions.length - 1
      ? "Lihat Keputusan 🏆"
      : "Soalan Seterusnya ➜";
}
function speakAnswer() {
  const SpeechRecognition =
    window.SpeechRecognition || window.webkitSpeechRecognition;
  if (!SpeechRecognition) {
    document.getElementById("voiceStatus").textContent =
```

```
    "Pengecaman suara tidak disokong oleh pelayar ini. " +
    "Sebut jawapan kepada guru, kemudian taip jawapan kamu.";
  return;
}
if (answered) return;
if (recognition) {
  recognition.abort();
}
recognition = new SpeechRecognition();
recognition.lang = "ms-MY";
recognition.interimResults = false;
recognition.maxAlternatives = 1;
document.getElementById("voiceStatus").textContent =
  "Mendengar... Sila sebut nama alatan dengan jelas.";
recognition.onresult = function(event) {
  const spoken = event.results[0][0].transcript;
  document.getElementById("answer").value = spoken;
  document.getElementById("voiceStatus").textContent =
    `Jawapan yang didengar: "${spoken}". Sila semak sebelum menghantar.`;
};
recognition.onerror = function() {
  document.getElementById("voiceStatus").textContent =
    "Suara tidak dapat dikenal pasti. Cuba lagi atau taip jawapan.";
};
```

```
  recognition.onend = function() {
    recognition = null;
  };
  recognition.start();
}
function nextQuestion() {
  if (currentQuestion < questions.length - 1) {
    currentQuestion++;
    showQuestion();
  } else {
    showResult();
  }
}
function showResult() {
  document.getElementById("quiz").classList.add("hidden");
  document.getElementById("result").classList.remove("hidden");
  const percentage = Math.round((score / questions.length) * 100);
  document.getElementById("progressBar").style.width = "100%";
  document.getElementById("finalScore").textContent =
    `${score} / ${questions.length} markah (${percentage}%)`;
  let message;
  if (percentage === 100) {
    message = "Hebat! Kamu memang detektif keselamatan yang cemerlang!";
  } else if (percentage >= 60) {
```

```
      message = "Bagus! Teruskan usaha untuk mengenali lebih banyak alatan keselamatan.";
    } else {
      message = "Teruskan berlatih! Kamu pasti boleh menjadi lebih mahir.";
    }
    document.getElementById("finalMessage").textContent = message;
  }
  function restartQuiz() {
    startQuiz();
  }
 </script>
</body>
</html>
```